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Transforming Challenges into Success: The Synergy of Engineering and Cutting-Edge Technology for Delivering Best-In-Class Water Injector Well

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Abstract

Water injector wells often face persistent technical challenges, including drilling complexities, wellbore instability, and completion inefficiencies, leading to significant operational and financial impacts. This study highlights a transformative approach that integrates multidisciplinary engineering expertise with cutting-edge technology to address these challenges. By employing advanced data analysis, innovative tools, and engineering-driven solutions, the team successfully drilled and completed a record-setting water injector well, achieving operational excellence and setting new industry benchmarks.

During well program preparation, the engineering team identified critical challenges, particularly the mother bore integrity, which resulted in deviations from the planned completion process. Analysis of offset wells revealed recurring issues linked to high-stress zones in the region. A multidisciplinary team conducted an in-depth stress map interpretation in collaboration with geological experts, confirming the adverse impact of these stress zones on wellbore stability and formation integrity. To mitigate these challenges, the team optimized the well trajectory and incorporated an innovative anchor technology into the completion design, effectively addressing operational risks and minimizing potential cost losses during the drilling and completion phases.

Following a comprehensive analysis and interpretation by the engineering team, recommendations for optimizing the directional drilling and completion plans were reviewed and approved. The solutions focused on refining the well trajectory to minimize exposure to high-stress zones and prevent fluid loss incidents. As a result, the well was successfully drilled without complete loss events, unlike offset wells. To address operational challenges, the team implemented a groundbreaking lower completion technique, utilizing innovative anchor technology. The system's slip mechanism securely grips the formation with minimal penetration, reducing formation damage while providing a stable, hydraulically set platform to manage operational loads. This anchor system also stabilized the completion string, preventing packer movement and enhancing the reliability and lifespan of downhole components. The integrated engineering plan and execution effectively resolved drilling problems and tubing movement issues caused by injection forces, marking the best-in-class water injector well performance for this field in over a decade. The project culminated in a remarkable achievement, completing the well in just 24 days with zero non-productive

time. This success set a new benchmark for operational efficiency and technical excellence in water injector wells' operations.

This case study demonstrates the transformative potential of integrating engineering expertise with innovative technology to overcome conventional limitations in water injector well design. The application of advanced trajectory planning and the pioneering anchor system showcases a scalable solution for similar challenges in other technically demanding fields. By redefining standard practices, this project underscores the importance of multidisciplinary collaboration and cutting-edge tools in achieving record-breaking performance and operational excellence.

Key Words: Water Injector Well, Multidisciplinary Engineering, Stress Mapping, Anchor Technology Innovation, Record-Breaking Performance, Operational Efficiency, Cost Optimization

Introduction

Efficient and reliable water injection is a fundamental aspect of reservoir pressure maintenance and improved oil recovery (IOR) strategies employed globally. As hydrocarbon reservoirs mature and become more complex, and as exploration and production activities extend into challenging geological environments, the design and construction of water injector wells are increasingly encountering significant technical hurdles (Serry et al., 2024). These challenges can substantially impact operational efficiency, the long-term integrity and longevity of the wellbore and completion, and ultimately, the economic viability of the overall field development project. Commonly faced difficulties in drilling and completing water injector wells include navigating through complex subsurface formations characterized by varying lithologies and properties, maintaining wellbore stability in regions with high in-situ stress anisotropy or natural fracture networks, mitigating severe to complete fluid loss during critical drilling and cementing operations, ensuring effective and sustained zonal isolation to prevent crossflow between different geological layers, and managing substantial axial and radial loads imposed on the completion string due to the dynamic conditions of water injection, including pressure cycles, temperature variations, and formation compaction over time (Rachapudi et al., 2018). Addressing these multifaceted and interconnected challenges effectively requires a fundamental shift from traditional, often reactive, approaches to a proactive, integrated strategy. This strategy must be deeply rooted in advanced engineering principles, a comprehensive understanding of the subsurface environment informed by robust data analysis, and the strategic deployment of innovative technological solutions.

A thorough understanding of the subsurface environment, with a particular focus on the in-situ stress regime and formation geomechanical properties, is an indispensable prerequisite for predicting and effectively mitigating wellbore instability issues during drilling and ensuring the long-term integrity of the completion. This section provides a detailed technical description of the geological context of the field and the prevailing in-situ stress conditions that exerted a significant influence on both the planning and the successful execution of the water injector well. Understanding the formation characteristics, including their strength, elasticity (Young's Modulus and Poisson's Ratio), and potential reactivity with drilling fluids, was critical for predicting wellbore behavior and designing appropriate mitigation strategies.

The presence of high-stress zones within the geological formations presented significant geomechanical challenges during well construction operations. The intricate interaction between the prevailing in-situ stress field, the orientation of the wellbore trajectory relative to these stresses, and the hydrostatic and dynamic pressures exerted by the drilling fluid can induce stress concentrations around the wellbore that exceed the mechanical strength of the surrounding rock, ultimately leading to wellbore failure. A primary mode of failure is the formation of wellbore breakouts. This occurs when the tangential compressive stress acting circumferentially around the wellbore exceeds the rock's unconfined compressive strength. The resulting spalling or fracturing of the wellbore wall leads to elliptical enlargements, typically forming in the direction of the minimum horizontal stress in a typical stress regime. Elevated differential horizontal

stress significantly magnifies the magnitude of this tangential compressive stress, substantially increasing the likelihood of breakout development, particularly when drilling wells perpendicular or at high angles to the maximum horizontal stress. Conversely, if the tangential tensile stress around the wellbore surpasses the rock's tensile strength, which is inherently low for most reservoir rocks, drilling-induced tensile fractures can initiate and propagate outwards from the wellbore. These fractures commonly orient themselves parallel to the direction of σ_H . High drilling fluid pressures (mud weight) directly contribute to increasing the tensile stresses around the wellbore, thereby exacerbating the potential for inducing such fractures. These wellbore failure mechanisms directly impact the operational window for drilling fluid density, known as the mud weight window. This window represents the range of mud weights that can be safely employed to maintain wellbore stability without causing either compressive failure (breakouts) or tensile failure (induced fractures). The lower bound of this window is defined by the formation pore pressure, while the upper bound is typically determined by the fracture gradient.

In high-stress zones, the increased magnitudes of both compressive and tensile stresses that can be induced around the wellbore effectively narrow this mud weight window. This technical constraint makes it considerably more challenging to select and maintain a drilling fluid density that is sufficiently high to prevent breakouts but simultaneously low enough to avoid inducing tensile fractures and the associated risk of fluid loss. These stress-induced wellbore failure phenomena and the resulting narrow mud weight window directly translate into significant operational problems, including tight hole conditions caused by wellbore swelling or caving that restrict drill string movement, stuck pipe incidents, particularly due to differential sticking in permeable intervals adjacent to breakouts, which are a major source of non-productive time (NPT), excessive cuttings generation from breakout spalling, leading to difficulties in hole cleaning and the potential for packed hole conditions, and fluid loss into drilling-induced tensile fractures, resulting in increased drilling fluid costs, reduced hydrostatic control, and potential reservoir damage if loss circulation materials (LCM) invade permeable zones. These interconnected geomechanical challenges, intrinsically linked to the severity of the high-stress regime, historically contributed to significant increases in NPT and overall well construction costs in offset water injector wells within the field, unequivocally highlighting the critical need for a proactive and technically robust engineering approach to mitigate these risks.

This paper presents a detailed case study demonstrating how a multidisciplinary engineering team, by applying sophisticated geomechanical analysis, optimizing well trajectory based on stress regime, and deploying a cutting-edge anchor system as a fundamental component of the completion design, successfully overcame persistent challenges to deliver a best-in-class water injector well. The project achieved record-breaking performance in terms of operational efficiency and near-zero non-productive time (NPT), establishing a new benchmark for similar operations in the field and providing a scalable model for challenging high-stress environments globally.

Methodology and Integrated Engineering Design

The project's paradigm shift towards success was fundamentally driven by a proactive and integrated engineering methodology that systematically addressed the identified geological and geomechanical challenges during the critical planning and design phases. This approach emphasized data-driven decision-making, cross-disciplinary collaboration, and the strategic selection and application of innovative technology as presented in [Figure 1](#).

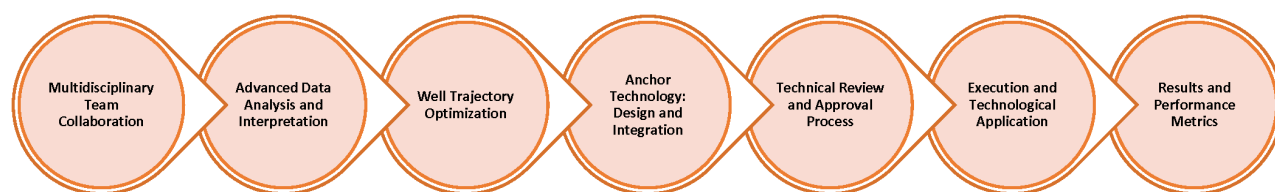


Figure 1—Technical Workflow and Engineering Methodology

Multidisciplinary Team Collaboration

The success of this complex well construction project was fundamentally underpinned by the formation and effective operation of a multidisciplinary project team from its inception. This integrated team comprised technical experts representing key engineering and scientific disciplines essential for addressing the challenges of drilling and completing wells in high-stress environments. The team included drilling engineers, whose responsibilities encompassed comprehensive well planning, meticulous trajectory design based on geological and geomechanical constraints, selection of appropriate bottom hole assemblies (BHAs) and drilling tools, optimization of drilling hydraulics for efficient hole cleaning and pressure management, and the development of detailed operational execution procedures. Completion engineers focused on designing the complex completion string, including strategies for achieving robust zonal isolation, selecting the appropriate packer types and setting mechanisms, designing the inflow control devices, and developing the detailed installation procedures for the lower completion and associated equipment. Reservoir engineers provided crucial insights into the characteristics of the target reservoir, including its petrophysical properties, fluid flow behavior under injection conditions, anticipated injection profiles, and the potential impact of wellbore construction activities on reservoir injectivity and long-term performance. Geologists contributed detailed knowledge of the regional stratigraphy, the specific lithological sequences encountered, and the presence and characteristics of any natural fracture networks or geological features that could influence wellbore stability or fluid flow. Geomechanics specialists were vital for their expertise in interpreting the in-situ stress regime through the analysis of wellbore failure data and other sources, analyzing the mechanical properties of the reservoir and overburden rocks, predicting wellbore stability for various trajectories and mud weight scenarios, and providing recommendations for optimal wellbore orientations and drilling fluid densities to mitigate stress-induced instability. This integrated team structure fostered a highly collaborative environment where technical insights, data interpretations, and operational constraints from each discipline were openly shared and considered interactively throughout the entire planning and design process. Regular cross-functional workshops and integrated data review sessions facilitated a holistic and shared understanding of the complex challenges at hand, enabling the team to develop technically sound, operationally feasible, and truly optimized solutions that addressed the well delivery process as a cohesive system, rather than optimizing individual components in isolation. This integrated approach stands in contrast to more traditional, sequential workflows, where the handoff between disciplines can sometimes lead to a less-than-optimal outcome due to a lack of comprehensive, shared understanding of the interdependencies between different phases and technical aspects of the project.

Advanced Data Analysis and Interpretation

The foundation of the integrated engineering design process was a thorough and advanced analysis of all available subsurface data, extending beyond routine review to employ sophisticated interpretation techniques for extracting maximum value from existing information. This involved a detailed wellbore image log analysis for stress and fracture characterization, where high-resolution wellbore image logs acquired from offset wells were meticulously examined to identify and determine the orientation of features indicative of stress-induced wellbore failure, specifically wellbore breakouts and drilling-induced tensile fractures. Quantitative analysis of the dimensions of these features, such as breakout width and depth and tensile fracture aperture, provided critical data for calibrating the magnitudes of the maximum and minimum

horizontal stresses. Statistical analysis of the observed orientations of both breakouts and induced fractures yielded reliable estimates of the orientation of the in-situ stress field. Furthermore, image logs facilitated the identification and characterization of natural fracture networks within the formations, assessing their orientation, aperture, and density, which is essential for understanding potential pathways for lost circulation and predicting wellbore stability in naturally fractured intervals.

Based on the interpreted stress orientations and calibrated magnitudes derived from wellbore failure analysis, leak-off tests (LOTs), and regional data, a detailed 3D in-situ stress model and a corresponding stress map generation and refinement process were conducted. This map provided a spatial representation of the principal stresses throughout the subsurface volume of interest and clearly highlighted the identified high-stress zones in three dimensions relative to potential and planned well paths. This precise and detailed mapping of the stress field was absolutely crucial for the subsequent phase of trajectory optimization.

Well Trajectory Optimization

The precise identification and comprehensive 3D mapping of the high-stress zones formed the technical basis for a data-driven approach to optimization of well trajectory based on stress regime. The primary objective of this optimization was to minimize the adverse mechanical impact that these stress concentrations would exert on the wellbore during drilling and completion. This was achieved by employing wellbore stability modeling as a function of trajectory. Geomechanical models predicted the distribution and magnitude of stresses induced around the wellbore as a function of the in-situ stress magnitudes and orientations, the planned wellbore trajectory (defined by inclination and azimuth), and the applied drilling fluid pressure (mud weight). Based on the predicted stresses around the wellbore and the rock's determined strength properties, the models were used for the prediction of wellbore failure risk.

Deviated wells intersecting the stress field at oblique angles can induce complex stress concentrations requiring careful analysis. To assess the impact of trajectory choices systematically, a comprehensive trajectory sensitivity analysis was conducted. This involved simulating wellbore stability for a range of potential well trajectories that could reach the reservoir target, evaluating trajectories with varying inclinations and azimuths as they traversed the identified high-stress zones. The analysis quantified the predicted risk of wellbore failure and the required mud weight window for each evaluated trajectory scenario. The final well trajectory was technically selected based on the results of the wellbore stability modeling and the trajectory sensitivity analysis. The chosen trajectory was the one that demonstrated the best predicted wellbore stability when passing through the high-stress zones, typically characterized by a wider and more manageable predicted stable mud weight window and minimized predicted risk of severe breakouts or induced fractures. This trajectory often represented a necessary compromise between the requirements for geomechanical stability and the objectives for reservoir accessibility. The technical basis and rationale for selecting this specific trajectory were thoroughly documented, demonstrating how it effectively mitigated the wellbore instability risks identified in the initial stress analysis.

Anchor Technology: Design and Integration

To address the critical challenges of maintaining wellbore stability during completion operations and mitigating the long-term issue of tubing movement induced by injection cycles, an innovative anchor technology was specifically selected and integrated into the lower completion design. This technology was chosen for its inherent capability to provide robust support within the wellbore and manage significant axial loads independently of the completion packer elements, a crucial feature in potentially challenging wellbore conditions encountered after drilling through high-stress zones. The anchor slip system ([Figure 2](#)) represents an innovative technology specifically designed to secure a liner within the open hole section of a well, providing crucial wellbore support and load management capabilities. A key technical feature of this system is its patented slip mechanism, engineered to grip the formation with strictly limited penetration. This controlled engagement minimizes potential damage to the formation during the setting process, a

significant advantage, particularly in sensitive or high-stress formations where excessive penetration could induce fractures or instability. Furthermore, the robust construction of the anchor slip ensures it provides a stable and reliable platform capable of handling the required axial and potentially radial loads anticipated during well operations. This system is typically hydraulically set and is often run at the bottom or toe of the completion string, functioning primarily as an anchor to prevent undesirable movement of the entire completion string during subsequent stimulation or production operations. This anchoring capability directly addresses the issue of tubing movement caused by pressure and temperature cycles, a common challenge that can lead to wear on open hole packers and compromise zonal isolation. By preventing packers from sliding up and down the wellbore, the anchor enhances the overall reliability and extends the lifespan of the packers and other downhole components. For applications requiring additional holding force, the anchor slip system is designed to be stacked, allowing multiple tools to be strategically placed along the completion string to increase the total load capacity. The system is particularly applicable in open-hole completions, high-pressure fracture stimulations where significant forces are generated, and horizontal wells where managing string weight and friction is critical. Technical features contributing to its effectiveness include an anti-preset mechanism to prevent premature activation of the tool during run-in, compliance with irregular hole geometries to ensure a secure grip even in less-than-perfect wellbores, and integrated centralization that helps maintain the completion string's position during dynamic operations. The design's inherent ability to prevent packer movement directly contributes to increased completion reliability, and its flexibility allows for placement anywhere in the completion string to optimize operational benefits.



Figure 2—Anchor Technology for Operational Excellence

Technical Review and Approval Process

The culmination of the integrated planning and design phase was a comprehensive and rigorous technical review and formal approval process, essential for validating the proposed well construction program before field execution. The multidisciplinary project team presented the finalized optimized directional drilling and completion plans to senior technical management and relevant stakeholders. This presentation included a detailed exposition of the initial geological and geomechanical analysis, the technical rationale behind the selected well trajectory optimized based on the in-situ stress regime, the design parameters of the drilling fluid system (specifically highlighting any modifications made to align with the optimized trajectory's requirements), and the detailed integration plan for the innovative anchor technology within the lower completion string. The review process involved a critical and in-depth evaluation across several key technical and operational dimensions. A primary focus was the risk assessment, which entailed a detailed technical assessment of the residual risks associated with the optimized plans. This included specifically evaluating the remaining wellbore stability risks when traversing the high-stress zones, despite trajectory optimization, and a thorough assessment of the reliable performance and potential failure modes of the innovative anchor technology under expected downhole conditions and loads. Technical feasibility was also rigorously verified, ensuring that all proposed operational procedures (e.g., specific drilling parameters, tripping speeds, circulation protocols, anchor setting procedures) and the application of the

selected technologies were technically achievable within the constraints of the available rig capabilities, personnel expertise, and the specific operational environment of the field. This evaluation confirmed that the integrated approach provided a more robust solution than a collection of individually optimized components. Based upon the successful demonstration of the technical soundness of the integrated plans, the effective mitigation of identified risks, and the projected favorable operational and economic benefits, the proposed well construction program received formal technical approval for execution.

Execution and Digital Platform Integration

The transition from meticulously engineered plans to successful field execution represents a critical phase where the effective collaboration between rigorous planning and advanced technology is fully realized. This phase involved the precise implementation of the optimized directional drilling and completion programs, an endeavor significantly enabled by real-time operational monitoring and the effective deployment of the selected innovative technologies.

Advanced engineering models were utilized to simulate the planned operation and predict potential outcomes, refining the plan for optimal performance and risk mitigation. Hydraulic modeling and simulation for the flow behavior of the drilling fluid in the annulus during tripping and circulation are essential for efficient hole cleaning and pressure management (Alharbi et al., 2025). Figure 3 illustrates hookload simulations and actual measurements during drilling operations for analyzing drag and friction. Multiple lines represent simulated hookloads under various friction factors (FF) as the solid black line represents the Hookload as a reference, while scattered points show the actual measured hookload. By comparing the actual hookload data points to the simulated curves, engineers can infer the effective friction factor experienced in the wellbore. If the actual points fall within the range of simulated curves, it suggests a reasonable friction model. Deviations could indicate changes in wellbore conditions, fluid properties, or sticking issues. This type of analysis is crucial for understanding wellbore friction, optimizing tripping speeds, and identifying potential problems during drilling and completion operations.

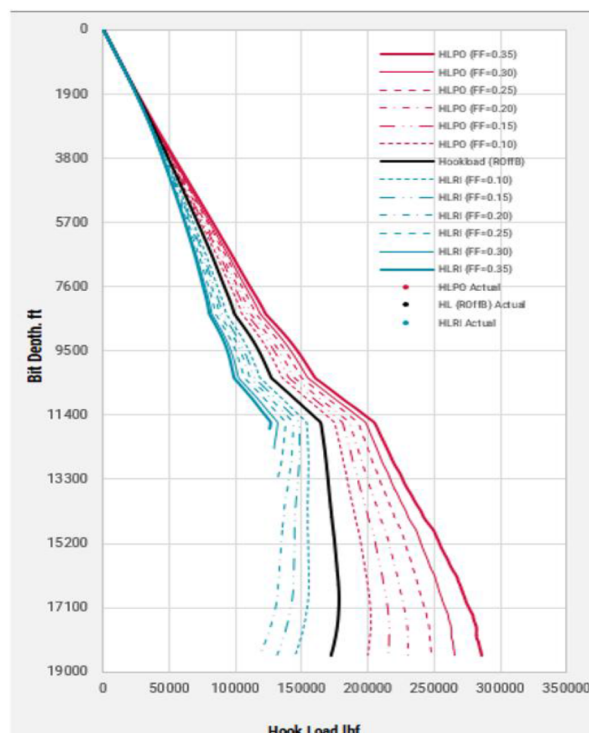


Figure 3—Simulated Hookloads Under Various Friction Factors

A digital platform served as the central nervous system for data management and real-time monitoring. Data acquisition and aggregation system was employed to capture high-frequency real-time data from all relevant surface sensors (hookload, torque, RPM, pump rate, standpipe pressure, flow in/out) and downhole sensors, when available, and stream the data into a unified database within the digital platform. Creating intuitive dashboards within the digital platform to provide real-time visibility of key performance indicators (KPIs), fluid properties, and operational parameters for the engineering and operations teams. Figure 4 represents a synthetic sample from the real-time drilling digital platform, providing real-time operational data and performance analysis. The operating parameters section displays current drilling metrics like bit depth, ROP, Flow In, Rotary RPM, Hole Depth, WOB, SPP, and Torque. The hookload broomstick plot visualizes hookload versus depth, showing simulated curves for various friction factors and comparing them to actual hookload data points. This allows for real-time estimation of the effective friction factor. Similarly, the torque broomstick plot shows simulated torque profiles against depth with actual torque data points. The model vs actual section uses gauges to display the percentage deviation between modeled and actual torque and hookload, providing a quick visual of how well the model is predicting real-time conditions. This dashboard integrates real-time data with predictive models, enabling proactive monitoring of wellbore friction, early detection of potential issues like sticking or packing off, and optimization of drilling parameters for improved efficiency and safety.

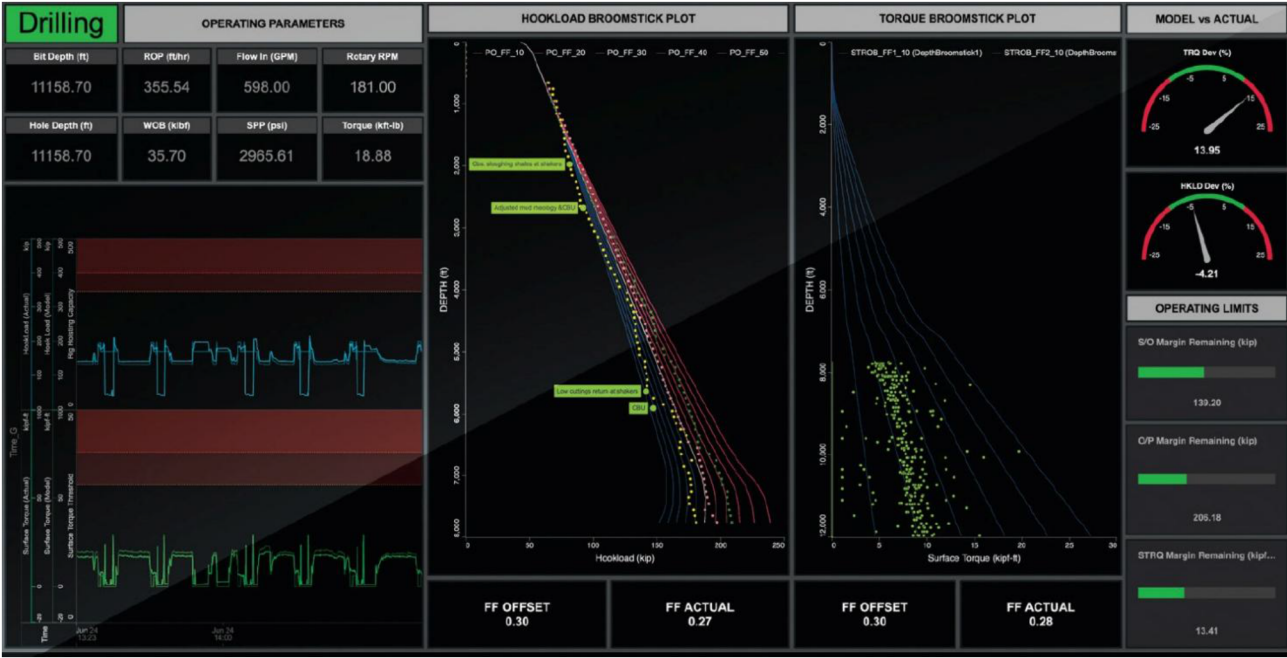


Figure 4—Drilling Digital Platform (Whitfield, 2022)

Results and Performance Metrics

The integrated engineering plan and its precise execution during the water injector well project yielded exceptional results, establishing new benchmarks for well performance within the field. These achievements are quantitatively supported by key performance indicators (KPIs) meticulously measured throughout and following the operational phases.

In terms of drilling performance metrics, the drilling phase demonstrated a significant step-change improvement when compared to historical data from offset wells in the same area. Crucially, the well was successfully drilled to total depth without experiencing severe or complete fluid loss events. This outcome directly contrasted with the historical performance of offset wells in the same geological intervals, resulting in substantial non-productive time (NPT). Furthermore, no stuck pipe incidents were encountered during

the entire drilling phase, providing strong evidence of effective wellbore stability management and efficient hole cleaning throughout the operation.

The entire process of drilling and completing the water injector well was successfully completed in a total duration of just 24 days. This represented a significant reduction in operational time, achieving approximately 30% reduction compared to the historical average for similar water injector wells in this field. The most significant achievement in this regard was the attainment of zero non-productive time (NPT) throughout the entire drilling and completion phases. This critical metric signifies that there were no unplanned delays caused by wellbore instability issues (such as stuck pipe or excessive reaming), fluid loss events requiring extensive remedial treatments, significant equipment failures causing major downtime, or technical problems encountered during the completion installation process.

Conclusions

This technical paper has demonstrated that a proactive, integrated engineering approach is essential for overcoming complex well construction challenges in high-stress environments. The case study of the best-in-class water injector well highlights the critical role of:

- **Advanced Geomechanical Analysis:** Detailed stress map interpretation and wellbore stability modeling are fundamental for understanding and predicting stress-induced wellbore failure.
- **Trajectory Optimization:** Strategically designing the well path based on stress regime can significantly mitigate drilling risks and improve wellbore stability.
- **Innovative Completion Technology:** The deployment of a novel anchor system capable of providing wellbore stabilization and managing injection-induced loads is crucial for ensuring long-term completion integrity and zonal isolation.
- **Multidisciplinary Integration:** Collaborative planning and execution across drilling, completions, reservoir, geology, and geomechanics are vital for developing and implementing holistic solutions.

The project's achievement of completing the well in 24 days with zero NPT provides compelling evidence of the operational and economic benefits of this integrated methodology. This success validates the technical soundness of the approach and sets a new benchmark for efficiency in challenging fields. The demonstrated ability to effectively manage stress-related wellbore instability and tubing movement through engineered design and innovative technology offers a scalable solution for similar technical challenges globally. This work underscores the importance of moving beyond reactive problem-solving towards proactive, data-driven engineering to achieve best-in-class well delivery performance and contribute to the sustainable development of hydrocarbon resources.

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